WORKED SOLUTION

STEP III 2015, Question 13

Probability and Statistics • Functions of Uniform Variables

STEP III 2015, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q13

The key idea

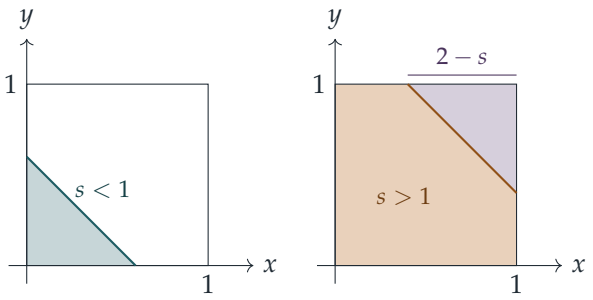
Because X and Y are independent and uniform on $[0,1]$, the pair (X,Y) is uniformly distributed over the unit square, so every probability about X and Y is simply an *area*. Events such as $X+Y \le s$ or $Y \le rX$ cut the square along a straight line, and the area on one side is a triangle or a square minus a triangle. After that, the distributions of $1/(X+Y)$ and $X/(X+Y)$ follow by rewriting their events in terms of the ones we have already measured.

Solution

Throughout, (X,Y) is uniform on the unit square $[0,1]^2$, which has area 1, so for any region A of the square, $\mathrm{P}((X,Y) \in A)$ equals the area of A.

Part (i)

Let $S = X+Y$, which takes values in $[0,2]$, and let $F(s) = \mathrm{P}(S \le s)$. The event $S \le s$ is the part of the square on or below the line $x+y=s$.



If $0 \le s \le 1$, the line meets the axes at $(s,0)$ and $(0,s)$, and the region below it is a right-angled isosceles triangle with legs s. If $1 \le s \le 2$, the region *above* the line

is a triangle in the corner $(1,1)$; its legs run from the line to the corner, of length $1-(s-1)=2-s$. Hence

$$F(s)=\mathrm{P}(X+Y\le s)=\begin{cases}0 & s<0,\\ \tfrac{1}{2}s^2 & 0\le s\le 1,\\ 1-\tfrac{1}{2}(2-s)^2 & 1\le s\le 2,\\ 1 & s>2.\end{cases}$$

The two middle formulas agree at $s=1$ (both give $\tfrac{1}{2}$), as they must.

Part (ii)

Let $T=1/S$. Since $0<S\le 2$ (the value $S=0$ has probability zero), T takes values in $[\tfrac{1}{2},\infty)$. For $t>0$, because $s\mapsto 1/s$ is decreasing on $s>0$,

$$\mathrm{P}(T\le t)=\mathrm{P}\big(S\ge \tfrac{1}{t}\big)=1-F\big(\tfrac{1}{t}\big),$$

using the fact that S is continuous, so $\mathrm{P}(S=1/t)=0$.

> **Ask yourself** Which formula for F do we use? It depends on where $1/t$ lies, not where t lies.

If $t\ge 1$ then $0<1/t\le 1$, so we use the first branch: $\mathrm{P}(T\le t)=1-\tfrac{1}{2}t^{-2}$. If $\tfrac{1}{2}\le t\le 1$ then $1\le 1/t\le 2$, so we use the second branch: $\mathrm{P}(T\le t)=\tfrac{1}{2}\big(2-t^{-1}\big)^2$. And $\mathrm{P}(T\le t)=0$ for $t<\tfrac{1}{2}$.

Differentiating gives the density. For $t\ge 1$,

$$f(t)=\frac{\mathrm{d}}{\mathrm{d}t}\Big(1-\tfrac{1}{2}t^{-2}\Big)=t^{-3},$$

and for $\tfrac{1}{2}\le t\le 1$, by the chain rule,

$$f(t)=\big(2-t^{-1}\big)\cdot t^{-2}=2t^{-2}-t^{-3},$$

with $f(t)=0$ otherwise. This is the required density.

For the expectation,

$$\begin{aligned}\mathrm{E}\Big(\frac{1}{X+Y}\Big)=\int_{1/2}^{\infty} t f(t)\,\mathrm{d}t &= \int_{1/2}^{1}\big(2t^{-1}-t^{-2}\big)\,\mathrm{d}t+\int_{1}^{\infty} t^{-2}\,\mathrm{d}t\\ &=\Big[2\ln t+t^{-1}\Big]_{1/2}^{1}+\Big[-t^{-1}\Big]_{1}^{\infty}\\ &=(0+1)-(-2\ln 2+2)+1\\ &=2\ln 2.\end{aligned}$$

The improper integral converges because t^{-2} is integrable on $[1,\infty)$.

Part (iii)

Let $R = Y/X$, which takes values in $[0, \infty)$ (the event $X = 0$ has probability zero). For $r \geq 0$, the event $R \leq r$ is $Y \leq rX$: the part of the square on or below the line $y = rx$ through the origin.

If $0 \leq r \leq 1$, this line leaves the square through the right-hand side at $(1, r)$, and the region below it is a triangle with base 1 and height r, area $\frac{1}{2}r$. If $r \geq 1$, the line leaves through the top at $(1/r, 1)$, and the region *above* it is a triangle with base $1/r$ along the top edge and height 1, area $\frac{1}{2r}$. So

$$H(r) = \mathrm{P}\Big(\frac{Y}{X} \leq r\Big) = \begin{cases} 0 & r < 0, \\ \frac{1}{2}r & 0 \leq r \leq 1, \\ 1 - \dfrac{1}{2r} & r \geq 1. \end{cases}$$

Now let $Z = \dfrac{X}{X+Y} = \dfrac{1}{1+R}$, which takes values in $[0, 1]$. For $0 < z \leq 1$, since $r \mapsto 1/(1+r)$ is decreasing on $r \geq 0$,

$$\mathrm{P}(Z \leq z) = \mathrm{P}\Big(1 + R \geq \frac{1}{z}\Big) = \mathrm{P}\Big(R \geq \frac{1-z}{z}\Big) = 1 - H\Big(\frac{1-z}{z}\Big).$$

Again the branch depends on where $(1-z)/z$ lies: it is at most 1 exactly when $z \geq \frac{1}{2}$.

- For $\frac{1}{2} \leq z \leq 1$: $\mathrm{P}(Z \leq z) = 1 - \dfrac{1-z}{2z} = \dfrac{3}{2} - \dfrac{1}{2z}$.
- For $0 < z \leq \frac{1}{2}$: $\mathrm{P}(Z \leq z) = 1 - \Big(1 - \dfrac{z}{2(1-z)}\Big) = \dfrac{z}{2(1-z)}$.

Both give $\frac{1}{2}$ at $z = \frac{1}{2}$. Differentiating (for the second, write $\frac{z}{2(1-z)} = \frac{1}{2}\big(\frac{1}{1-z} - 1\big)$), the density of Z is

$$g(z) = \begin{cases} \dfrac{1}{2(1-z)^2} & 0 \leq z \leq \frac{1}{2}, \\ \dfrac{1}{2z^2} & \frac{1}{2} \leq z \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Notice that $g(1-z) = g(z)$: the density is symmetric about $\frac{1}{2}$.

The expectation. We can write it down without integrating. The pairs (X, Y) and (Y, X) have the same distribution, since X and Y are independent with the same distribution. So $\dfrac{X}{X+Y}$ and $\dfrac{Y}{X+Y}$ have the same expectation. Their sum is exactly 1, so

$$\mathrm{E}\Big(\frac{X}{X+Y}\Big) = \frac{1}{2}.$$

Verification by integration.

$$\mathrm{E}(Z) = \int_0^{1/2} \frac{z}{2(1-z)^2}\,\mathrm{d}z + \int_{1/2}^{1} \frac{1}{2z}\,\mathrm{d}z.$$

The second integral is $\frac{1}{2}\big[\ln z\big]_{1/2}^{1} = \frac{1}{2}\ln 2$. In the first, substitute $u = 1 - z$, so $\mathrm{d}z = -\mathrm{d}u$, and $z = 0, \frac{1}{2}$ correspond to $u = 1, \frac{1}{2}$:

$$\begin{aligned}\int_0^{1/2} \frac{z}{2(1-z)^2}\,\mathrm{d}z &= \frac{1}{2}\int_{1/2}^{1} \frac{1-u}{u^2}\,\mathrm{d}u = \frac{1}{2}\Big[-\frac{1}{u} - \ln u\Big]_{1/2}^{1} \\ &= \frac{1}{2}\big(-1 - (-2 + \ln 2)\big) = \frac{1}{2}(1 - \ln 2).\end{aligned}$$

Adding, $\mathrm{E}(Z) = \frac{1}{2}(1 - \ln 2) + \frac{1}{2}\ln 2 = \frac{1}{2}$, confirming the value above.

Checks

- Each distribution function rises from 0 to 1 and is continuous where the branches meet: $F(1) = \frac{1}{2}$, $\mathrm{P}(T \le 1) = \frac{1}{2}$, $H(1) = \frac{1}{2}$, $\mathrm{P}(Z \le \frac{1}{2}) = \frac{1}{2}$. Each of these is $\frac{1}{2}$ by symmetry of the square about a diagonal.
- The density f integrates to 1: $\int_{1/2}^{1}(2t^{-2} - t^{-3})\,\mathrm{d}t = 2 - \frac{3}{2} = \frac{1}{2}$ and $\int_1^\infty t^{-3}\,\mathrm{d}t = \frac{1}{2}$.
- $2\ln 2 \approx 1.386$ lies between 1 and 2, as it should, and exceeds $1/\mathrm{E}(X+Y) = 1$, as Jensen's inequality for the convex function $1/s$ predicts. A direct numerical evaluation of $\iint (x+y)^{-1}\,\mathrm{d}x\,\mathrm{d}y$ over the square gives 1.3863. The distribution functions and $\mathrm{E}(Z) = \frac{1}{2}$ were also confirmed by computer simulation.

Common slips

- Choosing the branch of F or H according to t or z rather than according to $1/t$ or $(1-z)/z$, which swaps the two formulas.
- Forgetting that $1/s$ and $1/(1+r)$ are decreasing, so that the inequality must be reversed: $T \le t$ means $S \ge 1/t$.
- Getting the leg of the corner triangle in part (i) wrong: it is $2 - s$, not $s - 1$.

Beyond STEP

Turning a statement about (X, Y) into an area and then differentiating the distribution function is the general *distribution function method* for finding the law of a function of random variables. At university it is joined by the change-of-variables (Jacobian) formula for joint densities, which handles $X + Y$ and $X/(X+Y)$ together; in particular, the density of $X + Y$ here is the *convolution* of two uniform densities. Grimmett and Welsh, *Probability: An Introduction*, covers both methods, and Blitzstein and Hwang, *Introduction to Probability*, has many examples in the same spirit.

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